

VALUE AND MOMENTUM

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1. Overview of the Methodological Framework

This study constructs a systematic quantitative equity model for the Vietnamese stock market by integrating two well-established return premia: the value premium and the momentum premium. Drawing primarily from Asness, Moskowitz, and Pedersen (2013), who document the pervasiveness of value and momentum return premia across eight global asset classes, this methodology adapts their cross-sectional framework to the specific institutional, regulatory, and market microstructure characteristics of Vietnam.

The model encompasses three distinct portfolio strategies: (i) a value-only model rebalanced quarterly, (ii) a momentum-only model rebalanced weekly, and (iii) a combined model integrating both factors at weekly frequency. Each strategy constructs three portfolios through a within-sector tertile sort, labelled P1 (top third), P2 (middle third), and P3 (bottom third), following the spirit of Fama and French (1993) in their construction of size and book-to-market portfolios. Portfolio returns are computed using market capitalisation weighting to reflect the investable universe of a large institutional investor.

Several adaptations are made relative to the original framework of Asness et al. (2013) to accommodate Vietnamese market conditions. First, short selling is excluded from all portfolio constructions given the highly restricted and practically infeasible short-selling environment in Vietnam, rendering the model long-only throughout. Second, sector-specific value measures are employed, using the price-to-book (P/B) ratio for financial and banking stocks and the price-to-earnings (P/E) ratio for all other sectors, reflecting the differences in balance sheet structures across industries documented by Lakonishok, Shleifer, and Vishny (1994). Third, a 45-day publication lag is applied to all quarterly fundamental data to eliminate look-ahead bias, consistent with the approach of Fama and French (1992).

2. Data

2.1 Price Data

Daily adjusted closing prices and trading volumes are obtained for 301 stocks included in VNALL-INDEX, covering the period from January 2015 to May 2026. All price series are adjusted for corporate actions including stock splits, reverse splits, and rights issues.

The cleaned daily price matrix is converted into wide format with dates as rows and tickers as columns. Weekly prices are constructed using the last available Thursday closing price. This weekly Thursday series is used for the momentum signal and weekly holding-period returns. The script also creates monthly price and return matrices for reference, but the final value-model return computation is based on direct returns between quarterly rebalance dates rather than monthly aggregation.

2.2 Fundamental Data

Quarterly financial data including the price-to-earnings (P/E) ratio and price-to-book (P/B) ratio are obtained from FiinProX, covering the period from Q1 2021 to Q1 2026 for all 301 stocks in the VNALL-INDEX. The data are provided in wide format with quarterly observations as columns, and are reshaped into long format during processing. In line with the conservative data availability practices recommended by Fama and French (1992), a publication lag of 45 days is applied to all quarterly observations, meaning Q1 data (quarter ending March 31) is made available to the model from May 15 at the earliest. This procedure ensures that the model only uses fundamental data that was genuinely available to investors at the time of each portfolio formation date, thereby eliminating any look-ahead bias.

Any quarterly observation in which both the P/E and P/B values are missing is excluded from the dataset. Negative P/E values are excluded from the value signal computation, as they reflect loss-making periods in which the earnings yield is economically meaningless (Lakonishok et al., 1994). Additionally, P/E ratios exceeding 200 times are excluded as they typically indicate transitorily depressed earnings rather than genuine cheapness (Asness et al., 2013).

2.3 Market Capitalisation Data

Daily market capitalisation data are obtained from FiinProX for all stocks in the universe, covering the same sample period. Market capitalisation is defined as the product of the closing share price and the number of shares outstanding on the corresponding date. For each weekly portfolio formation date, the most recently available market capitalisation observation on or before that date is used, applying a forward-fill procedure analogous to that used for fundamental data. This approach prevents the use of future information in the construction of portfolio weights.

3. Signal Construction

3.1 Value Signal

The value signal is sector-specific. For stocks classified as Financials or Banks, the signal is book yield, defined as the inverse of the price-to-book ratio. For all other sectors, the signal is earnings yield, defined as the inverse of the price-to-earnings ratio. This distinction reflects the different economic interpretation of book value in financial institutions compared with non-financial firms, consistent with the motivation in Fama and French (1992, 1993) and Lakonishok, Shleifer and Vishny (1994).

$$x_{i,t}^V = \frac{1}{PB_{i,t}}, \quad \text{if } s(i) \in \{\text{Financials}, \text{Banks}\} \quad (3.1.1)$$

$$x_{i,t}^V = \frac{1}{PE_{i,t}}, \quad \text{if } s(i) \notin \{\text{Financials}, \text{Banks}\} \quad (3.1.2)$$

where $x_{i,t}^V$ denotes the value signal of stock i at date t , $PB_{i,t}$ is the price-to-book ratio, $PE_{i,t}$ is the price-to-earnings ratio, and $s(i)$ denotes the sector classification of stock i . For stocks classified as Financials or Banks, the value signal is book yield, $1/PB_{i,t}$. For all other sectors, the value signal is earnings yield, $1/PE_{i,t}$. A higher value signal therefore indicates a cheaper stock relative to its valuation anchor.

A higher value signal indicates a cheaper stock. At each quarterly value formation date, the model selects the most recently available valid value signal for each ticker whose safe date is less than or equal to the formation date. The value-only model then ranks stocks within sectors using this signal, with higher values assigned to P1.

3.2 Momentum Signal

The momentum signal is implemented as a cross-sectional return-based signal, but the construction differs across the weekly and monthly rebalancing specifications. This allows the analysis to compare whether momentum effects are stronger when portfolios are

updated frequently using short-horizon signals, or when portfolios are rebalanced less frequently using more persistent monthly formation signals.

For the weekly rebalancing specification, the price series is anchored to Thursday closing prices. At each weekly formation date t , the baseline one-week momentum signal for stock i is defined as the simple return from the previous Thursday close to the current Thursday close:

$$x_{i,t}^{M,W} = \frac{P_{i,t}^{Thu}}{P_{i,t-1}^{Thu}} - 1 \quad (3.2.1)$$

where $x_{i,t}^{M,W}$ denotes the momentum signal of stock i at weekly formation date t , $P_{i,t}^{Thu}$ is the Thursday closing price of stock i in the current week, and $P_{i,t-1}^{Thu}$ is the Thursday closing price of the same stock in the previous week.

In addition to the one-week signal, the weekly rebalancing framework also allows longer momentum formation horizons while maintaining weekly portfolio updates. More generally, the weekly momentum signal can be written as:

$$x_{i,t}^{M,W(a,b)} = \frac{P_{i,t-b}^{Thu}}{P_{i,t-a}^{Thu}} - 1, \quad a > b \quad (3.2.2)$$

where a and b define the beginning and ending points of the formation window in weekly units. In the empirical implementation, this framework is used to examine different momentum indicators, including short-horizon and longer-horizon momentum signals, while the portfolio is still rebalanced weekly.

For the monthly rebalancing specification, the momentum signal is constructed using closing prices on the last available trading date of each calendar month. Let $P_{i,t}^{ME}$ denote the closing price of stock i on the final trading day of month t . The monthly momentum signal is defined using two parameters, start month = s and end month = e , where $s > e$. The signal is computed as:

$$x_{i,t}^{M,M(s,e)} = \frac{P_{i,t-e}^{ME}}{P_{i,t-s}^{ME}} - 1 \quad (3.2.3)$$

where $x_{i,t}^{M,M(s,e)}$ denotes the monthly momentum signal, $P_{i,t-e}^{ME}$ is the price at the end of the formation window, and $P_{i,t-s}^{ME}$ is the price at the start of the formation window. In the main monthly specifications, $e = 0$, meaning that no skip month is imposed and the signal use information up to the most recent month-end trading date. For example, the one-month and one-quarter monthly momentum signals are defined as:

$$x_{i,t}^{M,M(1,0)} = \frac{P_{i,t}^{ME}}{P_{i,t-1}^{ME}} - 1 \quad (3.2.4)$$

$$x_{i,t}^{M,M(3,0)} = \frac{P_{i,t}^{ME}}{P_{i,t-3}^{ME}} - 1 \quad (3.2.5)$$

The monthly timing convention separates signal observation from portfolio implementation. The signal is measured after the close of the final trading day of month t . The portfolio is then rebalanced on the first available trading day after that month-end signal date, which corresponds to the beginning of the next monthly holding period. The portfolio is held until the next monthly rebalancing date. Therefore, the monthly holding-period return is measured from the beginning trading date of one month to the beginning trading date of the following month. This procedure ensures that the momentum signal is fully observable before portfolio formation and avoids look-ahead bias.

The use of momentum follows the cross-sectional momentum literature initiated by Jegadeesh and Titman (1993) and extended in broader asset-pricing settings by Asness, Moskowitz, and Pedersen (2013). However, the implementation is adapted to the structure of this study by considering both weekly and monthly rebalancing. The weekly specification captures short-horizon price continuation and allows frequent signal updating, while the monthly specification reduces turnover and tests whether more persistent momentum signals are more suitable for the Vietnamese equity market.

3.3 Weekly Value Signal for the Combined Model

The combined model requires both a value signal and a momentum signal at weekly frequency. Because the value signal is based on quarterly fundamentals, our methodology forward-fills the most recently valid quarterly value signal to each weekly date. A stock is eligible for the combined model only when it has a valid weekly momentum signal, a valid forward-filled value signal, and valid market capitalisation at the weekly formation date.

4. Portfolio Construction

4.1 Within-Sector Tertile Sort

All models use within-sector ranking to reduce unintended sector bets and to avoid ranking stocks across sectors using signals with different empirical distributions. At each formation date, stocks are grouped by ICB Level 1 sector and ranked by the relevant signal. The best floor ($n/3$) stocks in each sector are assigned to P1, the worst floor($n/3$) stocks are assigned to P3, and the remaining stocks are assigned to P2. This approach follows the spirit of sector-aware factor construction in the empirical asset-pricing literature, including Fama and French (1993), Asness, Moskowitz and Pedersen (2013), and Israel and Moskowitz (2012).

The minimum sector size is set to six stocks. If a sector has fewer than six valid stocks at a given formation date, all stocks in that sector are assigned to P2. This avoids constructing top and bottom tertiles from very small sector samples. Because the middle portfolio absorbs both remainder stocks and small-sector stocks, P2 is typically larger than P1 and P3.

4.2 Combined Value-Momentum Score

For the combined model, the value and momentum signals are first standardised within each sector using z-scores. This standardisation is necessary because the value and momentum signals are measured on different scales. It also ensures that the combined score is not mechanically dominated by the signal with the larger numerical dispersion.

Let $x_{i,t}^V$ denote the value signal of stock i at formation date t , and let $x_{i,t}^M$ denote the momentum signal. Let $s(i)$ denote the sector classification of stock i . For each signal $f \in \{V, M\}$, the within-sector z-score is computed as:

$$z_{i,t}^f = \frac{x_{i,t}^f - \mu_{s(i),t}^f}{\sigma_{s(i),t}^f} \quad (4.2.1)$$

where $\mu_{s(i),t}^f$ and $\sigma_{s(i),t}^f$ are the cross-sectional mean and standard deviation of signal f among stocks in the same sector as stock i at formation date t . If the sector-level standard deviation is zero or missing, the corresponding z-score is set to zero. Each z-score is then clipped to the interval $[-3,3]$ to reduce the influence of extreme observations.

The combined value-momentum score is computed as an equal-weighted average of the value and momentum z-scores:

$$C_{i,t} = 0.5z_{i,t}^V + 0.5z_{i,t}^M \quad (4.2.2)$$

where $C_{i,t}$ denotes the combined score of stock i at formation date t . Stocks are then ranked within each sector by $C_{i,t}$ and assigned to P1, P2, or P3 using the same within-sector tertile procedure described above. A higher combined score indicates a stock that is simultaneously cheaper on the value signal and stronger on the weekly momentum signal.

The equal weighting of the two standardised signals follows the empirical motivation in Asness, Moskowitz, and Pedersen (2013), who document that value and momentum contain complementary information across asset classes. In the Vietnamese market implementation, this structure allows the model to combine slow-moving valuation information with short-horizon price-trend information while maintaining sector comparability.

4.3 Sector-Replicated Market-Capitalisation Weighting

The final market-capitalisation-weighted portfolios do not use simple portfolio-level market-capitalisation weights. Instead, they apply a sector-replicated market-capitalisation weighting scheme. This approach first estimates each sector's weight in the full eligible investment universe, then reallocates capital across the sectors represented in the target portfolio according to those universe-level sector weights. Within each represented sector, stocks are weighted by their own market capitalisation.

This structure is intended to preserve the broad sector composition of the eligible universe while reducing unintended concentration in a small number of dominant large-cap stocks. This is particularly relevant in the Vietnamese equity market, where the investment

universe contains substantial heterogeneity in firm size, liquidity, sector composition, and trading activity.

Let $\mathcal{E}_t$ denote the eligible investment universe at formation date t , and let $\mathcal{P}_{k,t}$ denote the set of stocks assigned to portfolio k at date t , where $k \in \{P1, P2, P3\}$. Let $M_{i,t}$ denote the market capitalisation of stock i at formation date t , and let $s(i)$ denote the sector classification of stock i . The universe-level weight of sector s is first computed as:

$$W_{s,t}^U = \frac{\sum_{i \in \mathcal{E}_t, s(i)=s} M_{i,t}}{\sum_{i \in \mathcal{E}_t} M_{i,t}} \quad (4.2.3)$$

where $W_{s,t}^U$ is the market-capitalisation weight of sector s in the full eligible universe at formation date t .

For each target portfolio, the universe sector weights are then restricted to the sectors that are represented in that portfolio. Let $\mathcal{R}_t(k)$ denote the set of sectors represented in portfolio k at date t . The target sector weight for sector s in portfolio k is:

$$W_{s,k,t}^T = \frac{W_{s,t}^U}{\sum_{s' \in \mathcal{R}_t(k)} W_{s',t}^U}, \quad s \in \mathcal{R}_t(k) \quad (4.2.4)$$

where s' is a dummy index that runs over all sectors represented in portfolio k . This renormalisation ensures that the represented sector weights sum to one within each portfolio-date pair.

Within each represented sector, stocks assigned to the target portfolio are then weighted according to their market capitalisation relative to the total market capitalisation of all stocks in the same portfolio-sector group:

$$w_{i,k,t} = W_{s(i),k,t}^T \times \frac{M_{i,t}}{\sum_{j \in \mathcal{P}_{k,t}, s(j)=s(i)} M_{j,t}}, \quad i \in \mathcal{P}_{k,t} \quad (4.2.5)$$

where $w_{i,k,t}$ is the final portfolio weight of stock i in portfolio k at formation date t . The first component, $W_{s(i),k,t}^T$, determines the allocation to the stock's sector, while the

second component determines the stock's weight within that sector based on market capitalisation.

The resulting stock weights are normalised to sum to one for each portfolio-date pair:

$$\sum_{i \in \mathcal{P}_{k,t}} w_{i,k,t} = 1 \quad (4.2.6)$$

For the value-only and momentum-only market-capitalisation-weighted modules, this weighting procedure is applied separately to P1, P2, and P3. For the combined value-momentum market-capitalisation-weighted module, the same weighting procedure is applied to the P1 portfolio used in the reported performance summary, while the stock assignment output still records the P1, P2, and P3 classifications for each formation date.

The market capitalisation used in the weighting procedure is the most recently available observation on or before the relevant formation date. For the quarterly value model, this corresponds to the value portfolio construction date after applying the 45-day publication lag to the fundamental data. For the weekly momentum and combined models, this corresponds to the weekly formation date based on the Thursday closing price signal, with rebalancing assumed to occur on the following Friday. This timing convention avoids the use of future information in portfolio construction.

5. Trading Cost, Turnover, and Return Computation

5.1 Trading Cost Model

The final implementation applies explicit transaction costs in the market-capitalisation-weighted value, momentum, and combined modules. The configured trading cost assumptions are a 0.15% buy-side brokerage fee, a 0.15% sell-side brokerage fee, and a 0.10% sell-side tax. The implied round-trip cost is therefore 0.40%.

Cost component	Rate
Buy brokerage fee	0.15%
Sell brokerage fee	0.15%
Sell tax	0.10%
Total round-trip cost	0.40%

At each rebalance, the model distinguishes entering stocks, exiting stocks, and continuing holdings. Entering stocks are bought at their current target weights, and the buy fee is applied to those weights. Exiting stocks are sold at their drifted pre-rebalance weights, meaning the previous portfolio weights are first grown by the holding-period returns and then renormalised before applying the sell fee and sell tax. Continuing holdings incur cost only on the active rebalance trade: an increase in target weight receives the buy fee, while a decrease in target weight receives the sell fee plus sell tax.

$$\text{FeeDrag}_{k,t} = \text{BuyCost}_{k,t} + \text{SellCost}_{k,t} + \text{RebalanceCost}_{k,t} \quad (5.1.1)$$

$$\text{BuyCost}_{k,t} = \sum_{i \in B_{k,t}} w_{i,k,t} \times c^B \quad (5.1.2)$$

$$\text{SellCost}_{k,t} = \sum_{i \in S_{k,t}} w'_{i,k,t} \times (c^S + \tau^S) \quad (5.1.3)$$

$$\text{RebalanceCost}_{k,t} = \sum_{i \in H_{k,t}} [\max(w_{i,k,t} - w'_{i,k,t}, 0) c^B + \max(w'_{i,k,t} - w_{i,k,t}, 0) (c^S + \tau^S)] \quad (5.1.4)$$

where $B_{k,t}$ is the set of stocks entering portfolio k at formation date t , $S_{k,t}$ is the set of stocks exiting the portfolio, and $H_{k,t}$ is the set of stocks remaining in the portfolio. $w_{i,k,t}$ denotes the current target weight of stock i , while $w'_{i,k,t}$ denotes the drifted pre-rebalance weight based on the previous portfolio weights and realised returns since the prior rebalance. c^B is the buy-side brokerage fee, c^S is the sell-side brokerage fee, and τ^S is the sell-side transaction tax.

For the first rebalance of a portfolio, the model assumes the full portfolio is bought from cash, so turnover is set to 100% and the buy fee is applied to the full target portfolio weight.

5.2 Turnover

Turnover is computed as a weight-based measure rather than a stock-count measure in the market-capitalisation-weighted implementation. At each rebalance, the entering weight is the sum of target weights for newly added stocks, the exiting weight is the sum of drifted pre-rebalance weights for removed stocks, and the holding adjustment is the sum of absolute differences between target weights and drifted weights for stocks that remain in the portfolio.

$$\text{Turnover}_{k,t} = \frac{1}{2} \left(\sum_{i \in B_{k,t}} w_{i,k,t} + \sum_{i \in S_{k,t}} w'_{i,k,t} + \sum_{i \in H_{k,t}} |w_{i,k,t} - w'_{i,k,t}| \right) \quad (5.2.1)$$

The division by two prevents double-counting the buy and sell sides of the same reallocation. This measure better reflects actual traded portfolio capital than a simple count of changed tickers.

5.3 Gross and Net Return Computation

Gross return is computed as the market-capitalisation-weighted average of constituent stock returns over the relevant holding period. Stocks with missing holding-

period returns are excluded from the return calculation, and the remaining weights are effectively renormalised by dividing by the sum of weights with valid returns.

$$R_{k,t}^{gross} = \frac{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t} * r_{i,t}}{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t}} \quad (5.3.1)$$

$$R_{k,t}^{net} = R_{k,t}^{gross} - \text{FeeDrag}_{k,t} \quad (5.3.2)$$

where $R_{k,t}^{gross}$ is the gross return of portfolio k over holding period t , $R_{k,t}^{net}$ is the corresponding return after transaction costs, $w_{i,k,t}$ is the target portfolio weight of stock i , $r_{i,t}$ is the realised simple return of stock i over the holding period, and $\mathcal{V}_{k,t}$ denotes the subset of portfolio stocks with valid return observations. The denominator renormalises the portfolio weights when some constituent returns are missing.

For the value-only model, the holding-period return is computed directly from one quarterly safe-date rebalance to the next using the latest available daily prices on or before those dates. For the weekly-rebalancing momentum-only and combined models, the holding-period return is computed from one Thursday weekly price to the next Thursday weekly price. For the monthly-rebalancing momentum-only and combined models, the holding-period return is computed from one month-end trading date to the next month-end trading date. Annualised performance in the output is calculated arithmetically: average quarterly return multiplied by four for the value model, average weekly return multiplied by 52 for the weekly momentum and combined models and average monthly return multiplied by 12 for the monthly momentum and combined models. This is an arithmetic annualisation rather than a compounded annual growth rate.

6. Implementation Assumptions and Limitations

First, the framework is long-only. It does not implement short portfolios or long-short spread portfolios, even though the academic factor literature often evaluates value and momentum using long-short returns.

Second, the weekly momentum signal uses a one-week Thursday-to-Thursday return and assumes Friday rebalancing after the current Thursday close.

Finally, transaction costs are represented by fixed brokerage and tax rates. The model does not estimate bid-ask spreads, market impact, liquidity constraints, or execution slippage. The fee model is therefore a simplified implementation-cost approximation, consistent with the practical focus on turnover and fee drag emphasised by Frazzini, Israel and Moskowitz (2012).

7. Results and discussion

Table 1: Descriptive statistics of Net Portfolio Returns – Weekly Rebalancing

Panel A: One-week return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	4.0%	-3.7%	-5.6%	-1.7%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(0.47)	(-0.45)	(-0.62)	(-0.18)
Stdev	33.0%	24.0%	21.6%	19.5%	18.8%	20.6%	21.3%
Sharpe	0.06	0.39	0.82	0.19	-0.21	-0.29	-0.09
Alpha	-12.2%	-1.5%	7.8%	-8.6%	-16.2%	-18.7%	-12.5%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(-2.43)	(-5.78)	(-4.78)	(-2.69)
N	20	20	20	273	273	273	258

Correlation (Val, Mom) = -0.028

Panel B: Four-week return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	15.6%	4.4%	-3.0%	7.8%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(1.83)	(0.51)	(-0.31)	(0.83)
Stdev	33.0%	24.0%	21.6%	19.6%	19.9%	22.1%	21.0%
Sharpe	0.06	0.39	0.82	0.78	0.21	-0.15	0.36
Alpha	-12.2%	-1.5%	7.8%	3.1%	-8.8%	-16.8%	-2.9%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(0.83)	(-2.83)	(-3.62)	(-0.62)
N	20	20	20	273	273	273	258

Correlation (Val, Mom) = -0.060

Panel C: 12-week return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	17.8%	7.0%	2.9%	7.8%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(2.09)	(0.84)	(0.30)	(0.81)
Stdev	33.0%	24.0%	21.6%	19.5%	19.2%	22.3%	21.3%
Sharpe	0.06	0.39	0.82	0.89	0.35	0.12	0.35
Alpha	-12.2%	-1.5%	7.8%	6.5%	-5.0%	-10.8%	-3.0%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(1.03)	(-0.88)	(-1.59)	(-0.43)
N	20	20	20	273	273	273	258

Correlation (Val, Mom) = -0.091

Note: The table reports descriptive statistics for net portfolio returns after transaction fee. Mean, standard deviation, and alpha are annualised using arithmetic annualisation. Alpha is estimated as the intercept from a single-factor regression on VN-INDEX. N denotes the number of return observations. Correlation (Val, Mom) is the correlation between value and momentum z-scores

Table 1 reports the performance of the value-only, momentum-only, and 50/50 value-momentum portfolios under weekly rebalancing. It can easily be observed that the value portfolios show a clear failure of the expected value premium. Across all three panels, the value results are unchanged: the cheap portfolio, P1, produces an annualised mean return of only 2.3%, while the expensive portfolio, P3, generates 18.1%. More importantly, the alpha pattern is the opposite of what a successful value strategy would predict. P1 has a negative annualised alpha of -12.2%, with a t-statistic of -1.86, whereas P3 has a positive alpha of 7.8%, with a strong t-statistic of 3.91. This suggests that the cheap portfolio does not merely underperform in raw-return terms; it also underperforms after controlling for VN-INDEX exposure. By contrast, expensive stocks appear to earn positive abnormal returns. The evidence therefore points not only to the absence of a value premium, but potentially to an inverse value effect in this sample.

This result is important for interpretation. If low valuation multiples captured systematic underpricing, the cheap portfolio should have delivered positive alpha. Instead, the negative alpha of P1 suggests that cheapness may proxy for weak fundamentals, distress risk, poor liquidity, governance concerns, or low investor confidence. In the Vietnamese market, where disclosure quality, accounting reliability, and liquidity conditions vary substantially across firms, simple P/E or P/B-based value signals may fail to distinguish genuinely undervalued firms from structurally weak firms. The positive and statistically strong alpha of P3 also implies that expensive stocks may contain characteristics rewarded by investors, such as growth expectations, stronger quality, better liquidity, or higher institutional attention.

The momentum results under weekly rebalancing depend strongly on the momentum formation horizon. In Panel A, where momentum is measured using a one-week return, the strong momentum portfolio P1 earns a positive annualised mean return of 4.0%, but its alpha is negative at -8.6%, with a t-statistic of -2.43. This is a critical distinction: although P1 has a positive mean return, it does not generate positive abnormal performance relative to the VN-INDEX. P2 and P3 also have strongly negative alphas of -16.2% and -18.7%, respectively. Therefore, the one-week momentum signal does not appear to capture a robust abnormal-return effect. It may mostly reflect short-term noise, liquidity shocks, order-flow pressure, or temporary price movements that do not translate into market-adjusted outperformance.

The alpha evidence improves when the momentum signal is extended. In Panel B, the one-month momentum indicator produces a positive P1 alpha of 3.1%, although the t-statistic is only 0.83. Meanwhile, P2 and P3 have negative alphas of -8.8% and -16.8%. This produces a more coherent alpha ranking: strong momentum stocks perform better on a market-adjusted basis than weak momentum stocks. Panel C strengthens this pattern further. With the one-quarter momentum indicator, P1 alpha increases to 6.5%, while P2 and P3 remain negative at -5.0% and -10.8%. Although the P1 alpha t-statistic of 1.03 is still not statistically strong, the direction is economically meaningful. The weekly rebalancing results therefore suggest that momentum has more abnormal-return content when the signal is measured over one month or one quarter, rather than over one week.

The 50/50 value-momentum combination performs poorly under weekly rebalancing, especially when the momentum signal is very short-term. In Panel A, the combined portfolio has an annualised mean return of -1.7% and a negative alpha of -12.5%, with a t-statistic of -2.69. This indicates statistically meaningful underperformance after controlling for market exposure. In Panels B and C, the

combined portfolio improves as the momentum signal becomes longer, but alpha remains negative at -2.9% and -3.0%, respectively. This shows that the stronger medium-horizon momentum component partly offsets the weak value signal, but not enough to produce positive abnormal returns. The combined strategy is therefore constrained by the persistently negative alpha contribution from the value component.

Table 2: Descriptive statistics of Net Portfolio Returns – Monthly Rebalancing

Panel A: One-month return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	7.4%	6.5%	11.2%	3.1%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(0.83)	(0.71)	(1.14)	(0.28)
Stdev	33.0%	24.0%	21.6%	20.3%	20.6%	22.3%	23.8%
Sharpe	0.06	0.39	0.82	0.35	0.30	0.49	0.11
Alpha	-12.2%	-1.5%	7.8%	-4.9%	-6.1%	-2.1%	-8.6%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(-1.26)	(-1.62)	(-0.47)	(-1.71)
N	20	20	20	62	62	62	59

Correlation (Val, Mom) = -0.067

Panel B: Three-month return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	14.3%	10.5%	8.6%	6.5%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(1.51)	(1.14)	(0.90)	(0.58)
Stdev	33.0%	24.0%	21.6%	21.6%	20.9%	21.6%	25.0%
Sharpe	0.06	0.39	0.82	0.65	0.49	0.38	0.25
Alpha	-12.2%	-1.5%	7.8%	3.6%	-0.1%	-2.0%	-3.5%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(0.78)	(-0.03)	(-0.42)	(-0.67)
N	20	20	20	62	62	62	59

Correlation (Val, Mom) = -0.098

Panel C: 12-month return as indicator for Momentum							
	Value Portfolios			Momentum Portfolios			50/50 Combination
	P1	P2	P3	P1	P2	P3	P1
Mean	2.3%	9.6%	18.1%	16.0%	12.5%	4.2%	10.4%
<i>(t-stat)</i>	(0.16)	(0.90)	(1.87)	(1.61)	(1.33)	(0.44)	(0.90)
Stdev	33.0%	24.0%	21.6%	22.7%	21.4%	21.3%	25.8%
Sharpe	0.06	0.39	0.82	0.69	0.57	0.18	0.39
Alpha	-12.2%	-1.5%	7.8%	4.43%	1.5%	-5.8%	-0.09%
<i>(t-stat)</i>	(-1.86)	(-0.40)	(3.91)	(1.04)	(0.38)	(-1.09)	(-0.02)
N	20	20	20	62	62	62	59

Correlation (Val, Mom) = -0.047

Note: The table reports descriptive statistics for net portfolio returns after transaction fee. Mean, standard deviation, and alpha are annualised using arithmetic annualisation. Alpha is estimated as the intercept from a single-factor regression on VN-INDEX. N denotes the number of return observations. Correlation (Val, Mom) is the correlation between value and momentum z-scores

Table 2 repeats the analysis under monthly rebalancing. This specification is important because monthly rebalancing is more realistic from an implementation perspective, especially in a market where transaction costs, liquidity constraints, and trading frictions are material. Relative to weekly rebalancing, monthly rebalancing reduces portfolio turnover and makes the strategy less dependent on very precise short-term timing. The value results remain unchanged because the value strategy is based on the same quarterly signal structure, while the momentum and combined portfolios are evaluated using one-month, one-quarter, and one-year momentum indicators.

For monthly rebalanced momentum, the one-month momentum signal in Panel A is not successful. P1 earns an annualised mean return of 7.4%, but P3 earns a higher mean return of 11.2%. The alpha estimates confirm the weakness of this specification. P1 has a negative alpha of -4.9%, P2 has -6.1%, and P3 has -2.1%. Although the alphas are not as strongly negative as in the weekly one-week momentum case, the ranking is not consistent with a clean momentum premium. The strong momentum portfolio does not outperform on a market-adjusted basis. This suggests that a one-month momentum signal may be too short or too noisy when the portfolio is held for a full month.

The one-quarter momentum specification in Panel B produces a more favourable alpha pattern. P1 generates a positive alpha of 3.6%, while P2 is approximately zero at -0.1% and P3 is negative at -2.0%. The return ordering is also more consistent with the momentum hypothesis: P1 earns 14.3%, P2 earns 10.5%, and P3 earns 8.6%. However, the P1 alpha t-statistic is only 0.78, so the result is economically suggestive but not statistically strong. The evidence indicates that

quarterly momentum contains some abnormal-return information, but the sample does not provide enough precision to confirm a robust alpha effect.

Panel C provides the strongest monthly rebalancing momentum result. Using the one-year momentum indicator, P1 earns an annualised mean return of 16.0%, compared with 12.5% for P2 and 4.2% for P3. The alpha pattern is also the most coherent: P1 has a positive alpha of 4.43%, P2 has a positive alpha of 1.5%, and P3 has a negative alpha of -5.8%. This is consistent with the view that longer-horizon winners continue to outperform longer-horizon losers on a market-adjusted basis. Nevertheless, the P1 alpha t-statistic is only 1.04, meaning that the result remains statistically weak. The monthly results therefore support momentum more than value, but the evidence is still not conclusive.

The combined strategy under monthly rebalancing improves as the momentum horizon becomes longer, but it still does not produce convincing positive alpha. In Panel A, the 50/50 combination has a negative alpha of -8.6%, with a t-statistic of -1.71. In Panel B, alpha improves to -3.5%. In Panel C, where momentum is measured using one-year returns, the combined alpha is approximately -0.09%, with a t-statistic of -0.02. This is the best combined result, but it is still essentially zero abnormal performance. The improvement from Panel A to Panel C shows that longer-horizon momentum helps the combined strategy, but the persistent negative alpha of the value component prevents the combination from generating clear abnormal outperformance.

The most important overall insight is that the value signal fails in abnormal-return terms. The cheap portfolio has consistently negative alpha, while the expensive portfolio has consistently positive and statistically strong alpha. This is stronger evidence than the mean return comparison alone. It implies that the value strategy does not simply earn low returns because of market conditions; it underperforms after

controlling for market exposure. In this sample, cheap stocks appear to be cheap for fundamental reasons, while expensive stocks may reflect quality, growth, liquidity, or investor preference characteristics that are rewarded by the market.

The second main insight is that momentum performs better than value, but only when the signal horizon is sufficiently long. Very short-term momentum does not generate positive alpha. Under weekly rebalancing, the one-week momentum P1 portfolio has a negative alpha of -8.6%. Under monthly rebalancing, the one-month momentum P1 portfolio also has a negative alpha of -4.9%. By contrast, one-quarter and one-year momentum signals produce positive P1 alphas. The strongest evidence appears in the monthly one-year momentum specification, where P1 alpha reaches 4.43% and P3 alpha is -5.8%. This suggests that Vietnamese equity momentum is more likely to arise from medium- or long-horizon continuation than from short-term price movements.

The third insight is that the combined value-momentum strategy does not dominate the standalone momentum strategy. The combination improves when the momentum signal becomes longer, but its alpha remains negative or close to zero. This result is mainly caused by the value component. Since value P1 has a large negative alpha, adding it to momentum reduces the abnormal-return potential of the combined portfolio. The evidence therefore does not support a simple 50/50 combination in its current form. A better design would likely require either reducing the weight on value, conditioning value exposure on quality or liquidity screens, or using value only when it is not associated with distress.

The fourth insight concerns implementation. Monthly rebalancing appears more rational than weekly rebalancing for this setting. Weekly rebalancing with short-horizon momentum creates weak or negative alpha and is more vulnerable to transaction costs, liquidity shocks, and noisy price movements. Monthly rebalancing

with longer momentum signals produces cleaner alpha patterns and is more consistent with implementable portfolio management in an emerging market. However, even the better monthly momentum specifications are not statistically strong enough to claim a robust anomaly without caution.

Overall, the results suggest that the small Vietnamese equity sample that I have already used to examine does not support a conventional value premium, provides moderate support for medium- to long-horizon momentum, and offers limited support for a simple equal-weighted value-momentum combination. The central conclusion from the alpha estimates is that abnormal performance is concentrated in the momentum leg, while the value leg weakens portfolio performance. In Vietnam, market structure, retail trading dominance, liquidity constraints, short-selling limitations, and sector concentration may weaken the direct transferability of factor strategies documented in developed markets.

Table 3: Portfolio Turnover and Fee-Adjusted Returns – Weekly Rebalancing

	Model		Annualized gross return	Average turnover	Annualized fee drag	Annualized net return
Panel A	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (One-week)	P1	19.15%	73.07%	15.15%	4.00%
		P2	10.53%	68.46%	14.19%	-3.66%
		P3	10.54%	77.59%	16.09%	-5.57%
	50/50 Combination – P1		12.23%	67.24%	13.94%	-1.70%
Panel B	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (four -week)	P1	23.90%	39.92%	8.26%	15.63%
		P2	16.32%	57.46%	11.90%	4.41%
		P3	6.22%	44.47%	9.20%	-2.98%
	50/50 Combination – P1		15.78%	38.79%	8.02%	7.78%
Panel C	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (12-week)	P1	22.97%	25.11%	5.18%	17.79%
		P2	15.71%	42.05%	8.70%	7.01%
		P3	8.78%	28.37%	5.85%	2.94%
	50/50 Combination – P1		13.27%	26.58%	5.48%	7.78%

Table 1 shows that transaction costs have a material effect on weekly rebalanced strategies, particularly for momentum and the combined value-momentum portfolio. The value portfolios are less affected because they are rebalanced quarterly rather than weekly. As a result, annualised fee drag is low for the value portfolios, ranging from 0.28% for P3 to approximately 0.51%–0.52% for P1 and P2. This explains why the difference between gross and net returns is small for the value strategy. Therefore, the weak performance of the value P1 portfolio is not primarily driven by transaction costs. Instead, it reflects the poor return-generating ability of the value signal itself.

The weekly momentum portfolios show a very different pattern. In Panel A, where momentum is measured using one-week returns, turnover is extremely high across all three portfolios. P1 has average turnover of 73.07%, P2 has 68.46%, and P3 has 77.59%. This creates substantial annualised fee drag of 15.15%, 14.19%, and 16.09%, respectively. The impact is economically large. For example, P1 has a gross return of 19.15%, but after fees its net return falls to only 4.00%. P2 and P3 also have positive gross returns of around 10.5%, but both become negative after transaction costs. This indicates that the one-week momentum signal may generate frequent portfolio reshuffling, but much of the gross return is absorbed by trading costs. Under this specification, turnover is not merely an implementation detail; it fundamentally changes the economic viability of the strategy.

Panel B shows that using a one-month momentum indicator improves the trade-off between gross return and transaction costs, although fee drag remains high under weekly rebalancing. The P1 momentum portfolio earns a gross return of 23.90%, while annualised fee drag is 8.26%, leaving a net return of 15.63%. This is a much better result than the one-week momentum specification because the strategy generates stronger gross performance while requiring less costly rebalancing. However, P2 and P3 still experience considerable fee drag of 11.90% and 9.20%,

respectively. P3 is particularly affected: its gross return is 6.22%, but the fee drag exceeds the gross return, resulting in a net return of -2.98%. This suggests that even when the signal horizon is lengthened, weekly rebalancing remains costly, especially for portfolios with higher constituent instability.

Panel C provides the strongest weekly rebalancing result from a turnover-adjusted perspective. When momentum is measured using one-quarter returns, the P1 momentum portfolio has average turnover of only 25.11%, much lower than in Panels A and B. Annualised fee drag falls to 5.18%, while net return remains high at 17.79%. This means that the one-quarter signal not only produces strong gross performance, but also reduces unnecessary trading. The lower turnover is consistent with the idea that medium-horizon momentum is more persistent than one-week momentum. Because portfolio membership changes less aggressively, the strategy retains more of its gross return after transaction costs.

The combined 50/50 portfolio further illustrates the importance of turnover. In Panel A, the combined portfolio has a gross return of 12.23%, but average turnover of 67.24% produces annualised fee drag of 13.94%. Consequently, the net return becomes negative at -1.70%. This is a clear example of transaction costs fully eliminating gross profitability. In Panels B and C, the combined portfolio improves as the momentum signal becomes longer. With the one-month momentum signal, fee drag declines to 8.02% and net return rises to 7.78%. With the one-quarter momentum signal, fee drag falls further to 5.48%, while net return remains 7.78%. This suggests that the combined strategy becomes more implementable when the momentum component is based on a more stable signal. Nevertheless, even in the better specifications, transaction costs remain a meaningful drag on performance.

Table 4: Portfolio Turnover and Fee-Adjusted Returns – Monthly Rebalancing

	Model		Annualized gross return	Average turnover	Annualized fee drag	Annualized net return
Panel A	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (One-month)	P1	10.93%	73.97%	3.50%	7.43%
		P2	9.86%	71.95%	3.41%	6.46%
		P3	14.93%	79.02%	3.74%	11.19%
	50/50 Combination – P1		6.29%	68.47%	3.24%	3.24%
Panel B	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (3-month)	P1	16.45%	45.69%	2.14%	14.31%
		P2	13.48%	62.88%	2.97%	10.51%
		P3	11.03%	52.37%	2.47%	8.56%
	50/50 Combination – P1		8.73%	47.41%	2.22%	6.50%
Panel C	Value	P1	2.83%	35.00%	0.51%	2.31%
		P2	10.16%	35.62%	0.52%	9.63%
		P3	18.38%	20.4%	0.28%	18.09%
	Momentum (12-month)	P1	17.23%	26.05%	1.20%	16.03%
		P2	14.37%	40.70%	1.91%	12.47%
		P3	5.44%	27.66%	1.28%	4.16%
	50/50 Combination – P1		11.81%	30.06%	1.39%	10.42%

Table 2 shows that monthly rebalancing substantially reduces the annualised fee burden for momentum and combined strategies. This occurs because transaction costs are incurred less frequently than under weekly rebalancing. The value portfolio results remain unchanged because the value strategy continues to rebalance quarterly.

Panel A reports the monthly rebalanced strategy using a one-month momentum indicator. Although annualised fee drag is much lower than in the weekly rebalanced one-week momentum case, the return ordering is not favourable for momentum. P1 earns a gross return of 10.93% and a net return of 7.43%, while P3 earns a higher gross return of 14.93% and a higher net return of 11.19%. Average turnover remains high, with P1 at 73.97%, P2 at 71.95%, and P3 at 79.02%, but because rebalancing occurs monthly rather than weekly, annualised fee drag is limited to around 3.4%-3.7%. This indicates that monthly rebalancing reduces cost pressure, but the one-month momentum signal does not generate a clean pattern. The issue here is not primarily fees; it is the weak predictive power of the one-month signal under a monthly holding period.

Panel B shows a stronger result for the one-quarter momentum indicator. The P1 momentum portfolio has a gross return of 16.45%, average turnover of 45.69%, annualised fee drag of 2.14%, and a net return of 14.31%. P2 and P3 also have positive net returns of 10.51% and 8.56%, respectively, but P1 performs best. This produces a more coherent momentum ranking than Panel A. Importantly, fee drag is no longer large enough to dominate the strategy's economics. For P1, the gap between gross and net return is only slightly above two percentage points, meaning most of the gross return survives after transaction costs. This suggests that quarterly momentum is more suitable for monthly implementation than one-month momentum.

Panel C provides the most efficient monthly momentum specification. Using a one-year momentum indicator, P1 earns a gross return of 17.23% and a net return of

16.03%, with average turnover of only 26.05% and annualised fee drag of 1.20%. This is the best turnover-adjusted result among the monthly momentum portfolios. The P3 portfolio, by contrast, earns only 4.16% net. The relatively low turnover indicates that the one-year momentum signal produces more stable portfolio membership, which reduces implementation costs. The result is economically important because it shows that longer-horizon momentum not only improves return ordering but also materially improves cost efficiency.

In general, the turnover and fee results show that gross returns can be misleading when strategies involve frequent rebalancing, especially for short-horizon momentum. Therefore, the economic value of these strategies must be assessed using net returns rather than gross returns. Furthermore, it can be observed that transaction costs penalise unstable signals. The one-week momentum signal produces very high turnover and large fee drag, while longer momentum horizons generally reduce turnover and improve net performance. This pattern appears in both weekly and monthly rebalancing settings. The one-quarter and one-year momentum signals are more stable, leading to lower turnover and better preservation of gross returns. This supports the use of medium- or long-horizon momentum signals rather than very short-term price continuation.

The next insight is that monthly rebalancing is more appropriate for implementation in this market. Vietnam's equity market is subject to liquidity constraints, trading frictions, and potentially high price impact, especially outside the largest stocks. Although the backtest applies explicit transaction fees, real-world implementation costs may be even higher once bid-ask spreads and market impact are considered. Therefore, the reduction in fee drag from weekly to monthly rebalancing is economically meaningful. Monthly rebalancing provides a better compromise between signal responsiveness and cost control.

Finally, the value strategy's underperformance is not a turnover problem as the value portfolios have low annualised fee drag being rebalanced quarterly. Yet the cheap portfolio still produces weak net performance, while the expensive portfolio performs strongly. This means that refining the value signal is more important than reducing its trading frequency. The value strategy may need additional filters, such as profitability, quality, liquidity, or distress controls, before it can contribute positively to a combined strategy.

Overall, the turnover and fee analysis strengthens the conclusion that the most implementable strategy is not the highest gross-return strategy, but the strategy that preserves returns after trading costs. The best evidence favours longer-horizon momentum, especially the one-year momentum signal under monthly rebalancing. This specification produces high net returns, low annualised fee drag, and relatively moderate turnover. By contrast, short-horizon weekly momentum is too costly to implement efficiently, and the combined strategy only becomes attractive when the momentum component is sufficiently persistent and rebalancing frequency is reduced.

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